

## INTRODUCTION TO DBMS

Why DBMS: Human needs have increased tremendously. Now people are doing much more composite tasks than ever before. The society has become very complex. a person has to work with huge amount of information every day. In order to work with the enormous information, we must have a system where we can store, manipulate and share the information all over the world. It is one of the core reasons for introducing Database Management Systems (DBMS) as well as Relational Database Management Systems (RDBMS) now-a-days.

So, one thing is clear to us that we store and manipulate data / information into a database, where the database contains various types of tables for storing various types of data / information.

Data:

☐ Whatever we are inputting from the keyboard is known as Data.

It can also be called as RAWFACTS / FIGURES

☐ Data never provides any meaning for us

Information:

☐ Processed Data is known as Information

☐ Information always gives meaning for us

Database:

☐ Collection of information belongs to a particular topic (an organization) written in a predetermined manner stored at a particular place is called as database.

DBMS (Data Base Management System):

☐ It is software which is present inside the database, which can maintain and manage the data within the database.

Types of DBMS:

1) FMS / FDMS (File Management System /File Management Database System):

☐ This is first model released into the market in 1950"s. In this model there is always arranged in a continue stream of character (or) in a sequential fashion (or) manner.

Disadvantages:

☐ The Main disadvantage of this model is whenever we need to retrieve any data we have to start the searching from the beginning of the file so, it automatically leads to increases the searching time.

☐ Costly in maintains

☐ Required more man power

☐ There is no security

2) HMS/HDMS (Hierarchy Management System / Hierarchy Database Management System):

☐ This model was developed by IBM in 1960"s, When they developed a project called IMS (Information Management System)

☐ In this model data is always arranged in the form of a tree structure in different levels

☐ The top level can be called as root. The 2nd , 3rd can be called as parent and child levels respectively

☐ The main advantage of this model is we can easily retrieve the value without wasting much time

Drawback:

☐ Only one person can share the database simultaneously

☐ No security

3) NDBMS (Network Database Management System):

☐ This model was developed by IBM in 1969, when developing a project is called IMS (Information Management System)

☐ This model was developed on the basis of an Operating System called MULTICS (Multiplex Information Computing System)

☐ The main advantage of this model is more than one person can share the database concurrently (Simultaneously)

Disadvantage:

☐ There is no proper security for the centralized database

☐ Redundancy of the database is increased

☐ It occupies lot of memory and it leads to decrease system performance and increase the inconsistency.

4) RDMS (Relational Database Management System):

☐ This model was developed by a German scientist Mr. EF.CODD in 1970

☐ Here relation can be defined as commonness between objects these relations are classified into 3 types

- One to One relation
- One to Many relation / Many to One relation
- Many to Many relation

One – One relationship:

☐ In this relationship one object can have a relationship with another object

One - Many relationships:

☐ In this relationship one object can have a relationship with many objects

Many – Many relationship:

☐ In this relationship many vendors (or) many objects can have the relationship with many other objects

☐ All the above relationships can be called as “Degree of Relationships”

☐ This model was developed on the basis of a mathematical concept can be called as “Relation Algebra” (i.e. sets & Relations)

## Features of RDBMS

- ☐ In this model data should be stored in the form of tables
- ☐ A table can be defined as collection of rows & columns
- ☐ The horizontal lines are known as rows/ records / tuples
- ☐ The vertical lines are known as columns / fields / Attributes
- ☐ The intersection of rows & columns is known as cell
- ☐ A cell is a place where we can store our actual data
- ☐ The other name of table can be called as “Entity”
- ☐ It will provide high level security to database information
- ☐ Avoiding data redundancy problems
- ☐ Accessing the data from the table is not take much time
- ☐ When we define the column in the table user no need to follow any specific order

## INTRODUCTION TO SQL SERVER

SQL SERVER: SQL Server is an RDBMS product which was designed and developed by Microsoft Company.

SQL Server will provide Graphical User Interface (GUI) facilities it means that user can interact with the database using icons without remember any commands.

SQL Server will run only windows operating system i.e. it is a platform dependent.

The first version of SQL SERVER is 1.0 was released in 1989 and up to now 11.0 versions are available

Working with SQL SERVER: SQL Server is a collection of databases where database is a collection of various objects like Tables, Views, Procedures and Functions etc.

To work on SQL Server we use SQL Server Management Studio. It is a tool will provide Command Based Environment and GUI Based Environment to perform operations in SQL server. If we connect to server it shows a window with ....

- Server Type
- Server Name
- Server Authentication
- Username & Password

Server Type: SQL server contains five types of servers those are

- Database Engine: The Database Engine is the core service for storing, processing, and securing data (or) it is used to store, manage and to access the data from the database.
- Analysis Services (SSAS): It is used for data warehouse it will show the data in three dimensions (Rows, Columns and New dimension).
- Reporting Services (SSRS): It is a reporting tool used to generate reports in various formats such as creating interactive, tabular, graphical,

multidimensional, or XML-based data sources. Reports can include rich data visualization, including charts, maps etc.

- Integration Services(SSIS): It is used to convert tables from relational database to another relational database for e.g. If we want to convert SQL Server tables to ORACLE tables or My SQL tables then will be used.
- SQL Server Compact Edition: It is used to develop mobile application or mobile software.

Server Authentication: We have two types of authentications are

Windows Authentication: Windows Authentication work on the user admin and when we work on window authentication there is no required user name and password because operating system will generate User Id and Password by default.

SQL Server Authentication: SQL Server will work on the current user and when we work on SQL Server authentication then user should enter User Id and Password (These User ID and Password will give at the time of SQL Server installation).

Step To Connect To SQL SERVER:

Go to start → Go to programs → Go to Microsoft SQL Server 2008 R2/12 → Click on SQL server management studio → Click on connect button.

Object Explorer Window: This window contain Database, Security, Server Objects, Replication and Management options.

SQL Server contains two types of databases these are

- System Database: The system database include the following four databases
  - ☐ Master: It is used to manage system level information of SQL server.
  - ☐ Model: It is used as a template for all new creating databases in SQL Server.
  - ☐ Msdb: It is used to store the alerts and job information contains the SQL commands which are executed by user.
  - ☐ Tempdb: When ever SQL server is started tempdb will be created in SQL server. It is used to store temporary tables once we restart the server the tempdb database is destroyed.
- User Database: These databases are created and manage by the user for storing their objects like tables, views, procedure etc.

## Data Types in SQL Server

: A data type is an attribute that specifies what types of data enter by the user such as integer, character, decimal, date time etc.

- ☐ Integer Data Types
- ☐ Decimal Data Types
- ☐ Money (or) Currency Data Types
- ☐ Date & Time Data Types
- ☐ Character Data Types
- ☐ Binary Data Types

Integer data type: It will allows integer values only such as EID, SID etc

| Data Type | Range                                                   | Stored Memory |
|-----------|---------------------------------------------------------|---------------|
| Tiny Int  | 0-255                                                   | 1 byte        |
| SmallInt  | -32768 to 32767                                         | 2 bytes       |
| Int       | -2,147,483,648 to 2,147,483,647                         | 4 bytes       |
| Bigint    | -9,223,372,036,854,775,808 to 9,223,372,036,854,775,807 | 8 bytes       |

Decimal Data Types: These data types will allow only decimal numbers and it can divide into two types but both are same.

- Decimal (P,S) > P=Precision & D= Scale
- Numeric (P,S)

Precision: It allows the total number of decimal digits i.e. both left and right side of the decimal point. The default precision is 18 and maximum 38.

Ex of Precision: 3457.78543 Precision =9

Scale: It allows right side digits of decimal point only. The default value of scale is 0.

Ex of Scale: 3457.78543 Scale = 5

| Precision | Stored Memory |
|-----------|---------------|
| 1-9       | 5 bytes       |
| 10-19     | 9 bytes       |
| 20-28     | 13 bytes      |
| 29-38     | 17 bytes      |

Money Data Type: This data type will allows currency values and it contain two type these are

| Data Type   | Range                      | Stored Memory |
|-------------|----------------------------|---------------|
| Small money | -214,748,3648              |               |
| To          | 214,748,3647               | 4 bytes       |
| Money       | - 922,337,203,685,477,5808 |               |
| to          | 922,337,203,685,477,5807   | 8 bytes       |

Date and Time Data Type: These data types are define a particular date and time of the day.

Date: It defines date only the default format of date data type is "yy/mm/dd"

Time: It defines time of the day the default format is "hh/mm/ss.ms"

Date & Time: This will allows the both date and time of the day. "Yy/mm/dd hh/mm/ss.ms"

Character Data Types:

It allows to enter character values and these are classified into six types.

- Char (n): It is a fixed length data type, store the string

values in non-Unicode manner i.e. it will take 1 char per 1 byte.

The maximum length of char data type is from 1-8000 bytes

- Varchar (n/max): It is a variable length data type, store the string values in non-Unicode manner i.e. it will take 1 char per 1 byte.

The maximum length of Varchar data type is from 1-8000 bytes

- Text: It is same as Varchar(max) data type
- Nchar (n): It is a fixed length data type,

store the string values in Unicode manner i.e.

it will take 1 char per 2 bytes.

The maximum length of char data type is from 1-4000 bytes

- Nvarchar (n/max): It is a variable length data type, store the string values in Unicode manner i.e. it will take 1 char per 2 byte.

The maximum length of char data type is from 1-4000 bytes

- Ntext: It is same as Nvarchar (max) data type .

**Binary Data Type:** Binary data types are used to store images, videos and audio data. These can be divided into the following types

- Binary (n): It is a fixed length data type.

The maximum length of binary data type is 1-8000 bytes.

- Varbinary (n/max): It is a variable length data type. The maximum length of binary data type is 1-8000 bytes.
- Image: it is same as a Varbinary (max)

Note: Instead of text, Ntext and image data types are using Varchar (max), Nvarchar (max) and Varbinary (max) data types in latest versions of Microsoft SQL server.

## STRUCTURE QUERY LANGUAGE:

It is a non procedural language which is used to communicate with any database such as Oracle, sqlserver etc.

☐ This Language was developed by the German Scientist Mr. E.F.Codd in 1968

☐ ANSI (American National Standard Institute) approved this concept and in 1972 sql was released into the market



## Features of SQL:

- ❑ SQL is not a case sensitive language it means that all the commands of Sql are not case sensitive
- ❑ Every command of sql should ends with a semicolon (;)  
(It is exemption for SQL Server)
- ❑ SQL can be pronounced as Sequel  
(Structured English Query Language)
- ❑ SQL can be called as Common Language Interface, which is used to communicate with any type of database
- ❑ SQL can be called as NLI (Natural Language Interface).  
It means that all the SQL Commands are almost similar to normal English language
- ❑ Structured query language is mainly divided into 4 sub languages
  1. DDL (Data Definition Language)
  2. DML (Data Manipulation Language)
  3. TCL (Transaction Control Language)
  4. DCL(Data Control Language)

## SQL

DDL (DR.CAT) DML (SUDI) TCL (CRS) DCL

DROP      SELECT      COMMIT      GRANT  
SP\_RENAME      UPDATE      ROLLBACK      REVOKE  
CREATE      DELETE      SAVE TRANSACTION  
ALTER      INSERT  
TRUNCATE